

ActInf Textbook Group ~ Cohort 8 ~ Session 14

(Chapter 7) ~ 9/4/2025

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All right. Hello everyone. It's September 4th, aka 3 to the 2 power, 2 to the 2 power. Um 2025 is a square two and we're in chapter 7. So there's a few new learning resources and uh code examples that I have queued up. or we could take a big picture look at chapter 7. But let's just begin. If there's any section of the chapter or of anything that that comes to mind for people, let's just start there. So it could be a o overall remark on chapter 7 or its position in the book. um could be some section or or a question or an idea about about what chapter 7's doing or or what it's up to. Or I can give a sort of summary remark on what chapter 7 is is doing um and start to share some of these more recent resources too. But first, just anyone who has a comment or question they want to give can go for it or type it in the chat and I'll watch. Okay. Okay. Let's just zoom into a quick overview, but raise your hand or or write in the chat if anyone has a question. So again, the first half of the book is the epistemic and background content. The second half of the book is intended to be more like a workbook or a guide book getting us towards building specific generative models. Chapter 1 and 10, they're kind of like the book ends. That's the big picture and the theory review. Two and three give us the low and the high road to get to the generative models of active inference. And then five explores mammal neurobiology which is a really wellstudied system complex one but well studied. Then we head into the the back stretch. Chapter six is the recipe for designing active inference models. So those are kind of the ingredients and the processes that any given generative model, any given agent in environment model is going to have to address. So whether it's a rat in a T-maze or it's a motor circuit connected to an arm actuator, those are the kinds of questions that any given model is going to have to be addressing. Then we get into 789. So that's kind of the stretch that we're heading into right now. Seven and 8 go into more detail into the discrete time in seven and continuous time models in 8. That's kind of like a uh a little bit of a mitosis event relative to figure 4.3. So figure 4.3, we have these on the top discrete time

and on the bottom continuous time. We can talk more about their similarities and differences, but they're clearly laid out in figure 4.3 to show their structural similarity. And then chapter 7 and 8 kind of go into more detail about some features of discrete and continuous time models. And then chapter 9 talks about well what if we want to connect models to empirical data whether we want uh previous to chapter 9 it's more about just kind of instantiating a model and then generating synthetic data in chapter 9 the models get inverted where we have a situation that has already yielded data like we already have choice behavior from the mouse in the maze and now we're going to go back to parameterize some factor in the cognitive model. That's the big picture on the whole book. And seven is going to focus on the top figure here. Seven is going to build towards this model. So, here's where Oh, yes. Figure 4.3 page 70 in the book. Yeah. Or or or or just in the KOD as well. So again, chapter 4 where we're actually at that center point between the low and the high road. Chapter 2 coming up from the low road. Chapter 3 coming down from the high road. Chapter four in active inference city. And then we meet kind of the two types of entities there discrete and continuous time. Those get revisited in chapter 7 and 8. So that's where we are uh today. Um when it comes to chapter 7, well, I'll just share a few. Um this I I have done this over the last several days. Made this repo called start and did a short stream on it. This repo does a few things. First, it it hosts a bunch of links. Then it has a generative AI component to generate custom curriculum. So like for a given entity or audience, it will write a custom introductory curriculum. Not perfect, but a nice fun starting point. And then also it clones other repositories of applications. So like I've made a fork then of pmdp which kind of builds us up to what we get to in chapter 7. So just know that if we want to look at more of a code implementation of anything that we're about to go over, we can pop over to uh look at the code. Okay. So any comments or questions just on that sort of general situating of what chapter 7 is is doing Okay. So, um there's a lot to say about categoricalness and continuousness. And I'll start with one of on one of the the branches of PIMDP with this image of just a few um features of distributions. So >> hey Daniel, I'm still just seeing your the active inference textbook. Are you sharing your screen? >> Let me try again. Do you see this distributions? Okay. All right. So, um, two main distribution types that we're going to be talking a lot about are discrete distributions or the categorical distribution and then continuous distributions of which one type is a Gaussian or a normal. So categorical is where there are discrete differing options. So these are state spaces. This could be like temperature. We could be parsing up temperature into like it's cold, medium, hot, and there's just three options for temperature. Or we could have temperature be like a continuous variable

um in a in a normal distribution. and parameters have different values that characterize their shape. Um, chapter 7 and 8 are dealing with whether time is dealt with in a discrete or continuous fashion. Whether time is dealt with in continuous or discrete fashion, the state space could also be continuous or discrete. So, but time is kind of a special variable because of uh how it influences change in the dynamics of the model. And this is a common distinction that comes up in a lot of other modeling too. Um okay, so this chapter is going to focus on categorical variables in discrete time. So this is kind of the most categorical discrete possible quadrants where we're dealing with discrete clicks of time. Like time is just going to happen in these steps rather than being like a flow. And then the state spaces of the variables are also going to be these categorically different outcomes. There's going to be a sequence of examples which again have have um some light code presented in the appendix of the book in mat lab or you can look at the pmdp code um and it's going to build towards that figure 4.3 or as as it will come to be seen figure 7.3 is just that the top half of figure 4.3. three, we're going to get to this shape which we will um come to know and love in a stepbystep fashion. Any any comments or questions here? So the the first >> Sorry, can I ask a quick question? >> Yeah. >> Um tell me really quickly how to think about a discretet state space, you know, moving from one state to another state. And each of those discrete states is composed of a dynamic flow. Tell me real quick how to think about Yeah. Okay. So, let's in the discrete situation, it's like we're on a chessboard, like we're in a location that's discreetly different. Whereas the continuous case is like a field, like you have a GPS coordinate. So, it's still it still could be the same grid, but you're going to be continuously different. Then if you wanted to have a situation where there was at a higher level a discrete choice and then like within each cell you have a GPS field. So then you can say you know I'm in cell A2 at this higher discrete level and then within A2 I'm at 77.44. So that could be done with a nested model or there could be other ways but that's just sort of one one way to do it. Okay. So we have okay where where we're going is top half of figure 4.3 this partially observable marov decision process which is kind of like our this is going to be kind of like our Drosophila our fruitfly that we're going to work to in terms of summarizing the core cognitive features of perception and action selection. So the first example is just the downstairs. So this part is just the part that looks kind of like a hair comb or like a downward-facing letter E with a little handle. And what's happening here is there's a S. So a true latent state. This could be like the temperature in the room. And then at every latent state moment, every time subscript the current moment subscript towel or the past or the present or the past or the future with a negative 1 or plus one, it emits an observation through

A. So latent state is changing through time according to this B transition matrix. Every time step an observation is emitted like a thermometer reading is emitted from the latent state. And then we have D which is the initial prior on the latent state just to kick the chain off. But D is just used in the first time step and then for all later time steps like $T-1$ S_{T-1} plays the role of D to S_T . Okay, so this is just there's no action in this model yet. There's just the unfolding deterministic or probabilistic time evolution unfolding of this chain of S's. And then you also have a series of observations, but the observations aren't like linked to each other. they're only linked back to this main spine which is the time evolution of like the temperature in the room. But you do also have this sequence of thermometer readings. So this is a um hidden markov model because we have a hidden variable or latent variable s which is the one that's actually with a markovian property with this transition matrix but we get an observable at every time step. Um the example that that they use for this type of model is a music listening model. So it's like there's a sequence of notes, there's a true sequence of notes, and then there's a heard sequence of notes and then there's a transition matrix B which governs which notes are actually changing like in the score and then there are there are observations and beliefs which could be concordant or dissonant with how they are observed. Okay. So, it's just a pure perception example. It's basically just covering um those core downstairs ideas. How do you kick off the latent state with an initial prior? How do latent states evolve through time with a transition matrix that gives it a markovian property? How do latent states at each time emit an observation through the A matrix? And then I'll just add the sort of chapter 9esque idea is what if we run that a matrix backwards and instead of thinking about the latent state emitting a likely observation, we think about just having an observation and then pushing it back through the other way and asking okay given that observation what is the latent state app to be? Okay. So any just you know random comments or any questions on this perceptual downstairs part of the discrete time Last uh comment you made it has to do with uh the inversion. So that would be how you would get to your like your posterior or your recognition density. Exactly. That's like the so-called tale of two densities, >> right? >> Yeah. Is you can start from um the temperature of the room. Like let's just say this is our A matrix. So the 110th it it's written this way just so that there's integers there. But this could also be like 7.1.1 instead of this 110. Like the one10th distributes through the whole matrix. Um, so we could either um start with a real room temperature and then say what is it? Let's just say it's um cold, cool, warm, hot. So we could say we're going to be simulating this room that is cold. What temperature readings are we likely to get? Was it 70% likely that it's going to be cold, 10%

likely that it's cool? And so on. Or we could run it the other way and say we got this observation of it being very hot on the thermometer. It's 70% likely that the room is in that state. So that's the the tail two densities or the generative density which is going from the latent state to the observation that's like generative AI or from the observation going back to a latent state estimate. Okay, this this chapter comes at comes out one fast because that was the downstairs part. So now that's going to stay exactly the same. And now all that is added to get to a full active inference perception action joint distribution is these two variables up here. So now instead of B just being a single matrix that goes from from to to like the transition matrix of from sunny and cloudy days. Like it could just be that they stay the same 90% of the time and they change 10% of the time. So it's like $0.9 \cdot I$ a little square. just each time step you just apply that transition matrix and you end up staying in the current weather pattern 90% of the time. Now what's going to happen is there's a action selection variable or policy variable π and now there's going to now B is going to grow from being just a square matrix into being a tensor which is kind of like a collection of index cards. It could just be two index cards but they're all the same shape index card. And then what action selection is, it's like rifling through the index card file cabinet and saying which index card am I going to play to transition the latent state. So that is what action selection comes down to. Which transition matrix should I apply to the latent state time evolution? That's what π does. And then G is the expected free energy which is going to say how should I update how I select which index card to play. So we have π which is initially a prior on which action we're going to play and then you could just draw randomly from that uh policy prior or you could update the policy prior into a policy posterior using this expected free energybased procedure. which we'll talk more about so that you sample actions proportionally with their pragmatic and epistemic value. So that's kind of the key idea of the active inference components because on the downstairs we can do the basian updating the perceptual sense making all of that is happening with a variational free energy bounding surprise of observations um real-time inference even learning all those kinds of features don't require action selection at all and we get to full active inference when we're thinking about what the agent is doing as this joint distribution where it's able to first I mean where it's able to jointly update its beliefs about latent states and update its beliefs about actions Daniel. Um uh so I I understand the updating of the actions, but is it possible for those for there to be a like one of the index cards so to speak being uh gen being generative in the sense of I don't know it could be like random >> um a random matrix could still have a lot of structure. So, um, but yeah, like the, uh, one of the B , one of the B matrix slices could be like a

uniform distribution like just kind of grayed out or just sort of like SC depending on the situation like we can look at it in the T-mas and maybe see where that makes sense or not. But yeah, the the the slices of B represent beliefs about the causal consequences of selected actions. Now, there's kind of two layers of of of uh of challenge in the modeling there, which is summarized a little bit in figure 9.1. So, here's that figure 4.3, figure 7.3. This is the rat and the teenage that we're about to look at here. Were the experimentter in the lab making the cognitive model of the rat? And there's these kind of two uh two veils of unknowing. First, the rat could have a facious understanding of its actual situation. Second, we could have a facious model or a useless model of the rat. So the situation that that sort of seems to check all the boxes would be like we're we are properly calibrating a model to an agent that is skillfully engaging with the niche. That's kind of the paradigmatic case where we have a useful cognitive map of an agent that has a useful cognitive map. But the tools on our workbench allow us also to make the fantastical and the facious and the pedagogical models as well. So just by virtue of cobbling a model together using these components, it doesn't mean it fits the world well for an agent to have and it doesn't mean that we're going to fit the bird well just by building it. But but these are the components that help us get to those places. So like so and that the reason why I brought that up in response to like could there be kind of a random action? So it's like the agent environment question is like does the agent have an action it thinks it can select that would lead to a scrambling of the latent states in the environment. I mean maybe it's a reset button on the slot machine or something. So there could be a situation where that totally applies or maybe that could be nonsensical and then it would be would we be right in identifying that be for that organism. Um okay here um in the >> ask a quick question >> uh please >> who who is the policy known to um does the agent Yeah I guess that's my question >> yes great question it's a belief that the agent has about what it is apt to do. We don't need to take it into like the phenomenological. So it isn't necessarily that like it rises to a conscious awareness. It doesn't need to take on a sort of experiential baggage. We could talk about the policy of a bacteria to either run or tumble. We say it's 80% of the time it runs, 20% of the time it tumbles and it updates or it does action selection based on that. So we that doesn't need to take on the baggage like it's experiencing or having like a self-awareness of knowing run and tumble. But when we're doing this kind of modeling, it's like all of these variables are on board the agents. They're all part of >> you could take that on. Is that right? think that you could map that sort of in a hierarchical fashion if you will that there is a conscious awareness of pol discrete policy awareness if you want to call it that way. >> Yes.

>> Fair. Exactly. Yes. So, so at just at the first base level, all of the variables here, A, B, C, D, E, you know, F and G, basically π_i , those are the joint distributions. That's the packet, that's the Python class that you need to describe the agents. Then you could have a hierarchical model where it's like okay we're going to have a kind of supervening model or supervisory or a hierarchical model that's looking at π_i as an observation coming up a hierarchy and that's where people start to get into like um introspection and metacognition as awareness is like if a variable could be observed. I mean, observing an internal variable is metacognition. And then mental action is well, what if you what if that was a two-way street and observations of internal variables were arising and then mental action or attention or control were a descending policy of metacognitive variables on first order cognitive variables. Okay. Um, could I I have a question about terminology? Uh, I've I've come to interpret belief as uh what has been experienced which is is related to why I asked my question. You know in terms of creativity we often think of that again on a very high level as making new connections between existing uh connections. Uh but again the interpretation of belief is like what exists or what has been experienced. Which is why I asked the question can it be can the belief be something completely newly just um generated? I guess in a way you got at that at the random uh instantiation but again the terminology belief or or or prior it's it's something that's been experienced. It's like the default. Yeah. I I think there's a lot of pieces there. The prior which is to say before the next observation is like the calibration of the sensor before the next observation comes in. And then when it comes to novelty belief updating is going to stay within the rails. Like if it's a distribution that varies between zero and one you're going to move around within that zero to one space. And then there's this kind of like categorical novelty with like what if you what if the structure of the belief changes and that's kind of approached in structure learning where the actual like type of the variable can change or a new node could appear or connect connections like components could change in the graphical model which opens up like a a very massively combinatorial brick creative novelty. Whereas this is more like it's kind of like the plane is already built. It's in flight and like each of the flaps could be between zero and one. The engine can go to but like we're not going to be adding new flaps or taking flaps off. So it's kind of like on the rails within the support. That's that's the the term that comes up a little earlier in the book is support which is basically the X range that a statistical distribution covers. So those that's like the state of play for a given variable. So we got to the kind of key 7.3 which is this downstairs sensemaking component now with a policy selection variable that's picking which slice of B is going to be used in the

causal unfolding. Um, so instead of just here's the here's the um hot and cold that stays the same temperature 90% of the time and it flips 10% of the time. That's just a single 2x2 matrix. Now we have turn on the air conditioner, do nothing, turn on the heater and the air conditioner biases the temperature down and the heater biases it up and do nothing keeps it, you know, one and keeps it the same or you could have it it randomizes the the temperature, whatever. And then there's a procedure with expected free energy for how you update your policy prior into a policy posterior. So it's the same kind of general basian concept. This is why it is literally active inference because it's policy and action selection as an inference problem to update your policy prior belief which is what what I was apt to do before I updated beliefs about how it's apt to be. Then after I've updated how it's apt to be, what am I apt to do given what information I might receive from actions which is the epistemic value and how action might orient with my preferred observations. There's calibration to be done to actually finesse the exploit explore The special case is the purely pragmatic. This is the special cases that are described in figure 26. Here's expected free energy pragmatic value and epistemic value. So it's the other order. But here is epistemic is the first term, pragmatic is the second term. Here's the other way around. Just to be clear, if you only have pragmatic value, >> if you're showing the 2.6, we're not seeing it. >> Oh, it's kind it's like I updated Pop OS. Okay. 26. If you only have the pragmatic value term, you have pragmatic value or utility maximization which is basically reinforcement learning. You have a a value function. You have a utility function and then actions are selected that either you pick the best action or you draw from the posterior probability of actions proportionate to their expected utility return. Conversely, if you the other special case of expected free energy is if you say actually I equally prefer all outcomes and that connects to that kind of like equanimity type um phenomenology and you only have epistemic value. Epistemic value maximization leads to a noveltybased search where actions are selected solely upon their expected information gain. And then there's this finessing of how much and in what ways and with what switching strategy of which level of the model etc etc etc. Do you want to pursue pragmatic and/or epistemic ends? And if the situation is such that the policies that are pragmatic are also yielding epistemic value, then there's no conflict. So it doesn't really matter. But that finessing comes into play when there are policies that have where those vectors are not completely aligned. And in most situations you can imagine that they're probably not. But you can design a situation where they completely are. So in the general case there there's there's no there's not too much to say just that you could you could imagine those are like the state spaces of of different models and different information

possibilities. Now now do you see the textbook again? Okay. Okay. But what G does is it takes in the policy prior π_i . It scores each of the policies according to those two attributes. We can go into more detail with what the variables are if people want, but it scores policies according to that. Um, and then it reweights the policies likelihood of being selected based upon their score on that G . And then with your action posterior, your reweighted policy, you know what I'm apt to do now, now that I know what I know now, actions are either you take the best scoring policy or you sample distribution of policies. So that's the upstairs action selection process. So that's how the whole continuum from epistemic novelty oriented behavior novelty search on through purely reinforcement reward pragmatic utility that end of the pool and modulations in between are treated as part of this unified action inference problem. which gets composed onto a sensemaking perception problem without without the negative connotation of problem or puzzle. So that we have this downstairs sensemaking, upstairs action selection that represent those two fundamental inbound perceptual and then outbound action emitting aspects of things. 7.456 goes into more detail on the rat in a T-maze and uh PMDP has a great example of a T-maze and I've I've made some additional uh code methods if to to look through. 7.4.4 goes a little deeper into epistemic value in terms of giving a few different ways to to really zoom in on what what's happening with um so the rat basically here's its options. It starts in the starting position. There's only four positions in this whole scenario. There's the starting position. Then there's two ends of the maze. one of them has a preferred sensation like sugar and the other has nothing or a dispreferred sensation. Um you know so people have done the tests in the sort of reward seeking or in the uh punishment aversion setting or both and then the bottom position is a Q. So basically the rat can go to any of to make three choices. It could either go to the left, go to the right or but it doesn't know which one has the preferred observation. What it can do is it can go to the Q which it has already used associative learning in this situation to associate okay if it's an L that means that the good stuff is on the left. So, it's like there's there's, you know, there's two slot machines up ahead. One of them is a high yield, one of them is low yield, but I'm not exactly sure which one is high and low yield. So, I could just jump the gun and pick one or the other. Or I could ask this person who I believe is going to be a source of uncertainty reduction onto and then I'll go to which one I think is better. So like that kind of micro decision which is a situation where it's like and that kind of relates to like time preference and and delayed gratification in a sense. It's like if it were \$10 and \$1 between the two slot machines, but it's like but if I go to the queue, maybe the queue costs money to go to or may at the very least it's a time delay. So, how do we model that situation

where information gain is being traded off either temporally or in terms of uh some kind of currency with the possibility of rewards 7.1 is a little bit box 7.1 is A bit of a more background concept about about um entropy of distributions just where when it's uh between uh uh whether it's the inverse temperature or the temperature of course sets up whether it's going to be a higher or a lower value. But there's a knob that you can turn between something being flattened or being sharpened. It's kind of like the gain on a distribution. Uh 7.9 is just is a citation to dig into more. It's not really presented with the equations here. Um where there's four locations. So it's actually in that way quite similar to the T-maze because there's also four locations in the T-maze but here it's an eye cade visual foraging task. So the visual set I mean and this is like one of the amazing truth drops of the predictive mind which is that under healthy vision we have color and clarity in our whole visual field. Even though color and clarity is actually only a very small fraction at the center of our vision, it's just like one of the most most like mic drop. Like that's why we're perceiving belief, not processing inbound retinal photons. The retinal photons are just fine-tuning the generative model of what we believe to have color and clarity, even if we weren't looking at it. So the visual foraging is a really interesting example for that. Then 7.5 uh goes into a little bit about how learning can happen. So here we have that same seeing. Yes. See with a mind. Here we have the same downstairs hair comb. Here's D SS SS O and the same kind of upstairs action selection but there's some other variables added. We can talk more about it but that it relates to how each variable in the base POMDP. You could either just take it as fixed or you could take each of those variables like the A or the B or the D as drawn from a hyper prior distribution. So it's like in like if it were rainy or sunny, you could say D is 0.5. It's just 50/50 just that. Or you could say actually my prior on the weather is 0.5 plus or minus.2. So then across trials it'd be like okay now I draw and it's 60/40. Now I draw and it's 40/60 and then it's 55/45. So then across many many trials you would end up getting this normal distribution around 50/50 but it wouldn't exactly always be 50/50. So that's like the idea of using Okay, this one Mark. Yeah. Yeah. So here's here's the D that we were working with before like 50/50 sunny cloudy but it may be drawn from another distribution that could be like so exact details are going to depend on what shape the distributions are but basically you can see you could have instead of just like the 50/50 is like a spike and that itself could be a distribution or like the preference values it you know it could be I I prefer the le prefer the sugar versus the nothing five to one you say or I prefer it 5:1 plus or minus one so then on the next trial it could be 6 and.5 or something like that. So that's what a hyper prior is is it it it like the bucks stops somewhere. There is a level of

the model where you just say okay it's just this is just the variable but you can always in Bayesian statistics add another variable which that one below it is sampled from. So it seems to be bypassing that sort of expected free energy. I guess that's where I'm kind of getting hung up here. It's having it's exerting its own influence >> and so it's sort of like the given. Am I I don't know if I'm saying making sense there. It's elaborating on the previous POMDP. It's giving it the possibility of learning of like multi-trial learning. It doesn't really bypass. It's just in a separate it's like this is kind of our city. So, and the edges are the information flows. So it's like again if it's 50/50 prior on the weather, we could have a whole causal model that leads us to come to believe that in this situation it's it's 8515 for the weather and then that would just be used as the prior on S and then everything else could go down the road exactly the same. So a lot of times the Markov blanket is talked about as if it were the one that is partitioning the agent from the world and that is an important blanket to consider. But also just more generically and uh less um or more philosophically spicy depending on how you think about it. Any intervening variable blankets off any other like the influence of D on S subt only is acting through S to minus one or the influence of this A learning variable little A on O you know on the relationship between S and O is only acting through here. So you can add whole branches that are actually Markov blanketed off just because of topological partitioning. So that kind of speaks to the composability of the model is there's different ways to swap out components like a matrix can get very large and in those situations that whole component can get swapped out with a neural network. So, we're kind of learning the matrix-based formulation, which is the kind of simplest, most analytically well-defined, most interpretable, most index card, kind of back of the sheet. But if the matrix were 10,000 by 10,000, there could be a situation where you just swap it out with a neural network. D could be its own neural network for that matter. >> Yeah. Sort of making a synaptic connection to >> an existing. >> Okay. >> Yeah. Like like so let's say the S is going to be whether it's cloudy or sunny. So it's just it's just going to be give us, you know, 8020. But we could to kick off the chain, we could um download all the weather reports um download all this data, send it to an LLM and then say LLM returning only two numbers that sum to one. What is the probability that you believe it's raining? And it will chug chug chug chug and then come back 0.9.1. So that would be like the LLM playing the functional role of D emitting the shape of the variable that we need to kick off this whole chain. But we're just going to write D as.9 in this kind of pure minimal matrix form. But like these are the kinds of uh debates that people have which was like well if we replace every variable with a neural

network like is it still active inference or does it have to be just this kind of matrix form like analytically closed um style 7.2 two talks a little bit about this idea of conjugate priors which is basically what does little D have to look like so that big D is the shape that we need. So that's a little bit about learning and like that comes into this learning by counting idea. So it's like that 0.5 we can think about it. It has like a sort of parent distribution that's like 1 one and then 1 one goes to 0.5 and then we see a sunny day. Now it's 2:1. Now D goes to $\frac{2}{3}$ $\frac{1}{3}$ and then we see now it's 3 1 and then we update D to 75.25. So that's what the conjugate distribution does in the learning by counting situation is it just runs a tally and then it uses the ratio of the tally as the prior for the next. And that's also related to like the urn example which is like you reach into an urn and you pull out a ball. It's blue. Okay. So now you believe that it's 100% blue in the urn. Then you pull out a purple one. Now you believe it's 50% blue, 50% purple. Now you pull out a second blue one. Now it's $\frac{2}{3}$ blue, $\frac{1}{3}$ purple. And you keep on drawing. And then you could come to a sort of um understanding of when you were asymptotically sampling. So learning by counting example with a citation structure learning just sort of alluded to but that's this idea that Bayesian graphs that have different structural features can be compared in terms of their uh their posterior evidence. And then this hierarchical modeling idea, another example involving uh uh multiple time scales and that's the chapter. So, it's kind of this one's kind of like a roller coaster because it sets up the two real main dishes with that downstairs sensemaking. The upstairs action selection goes in deep to the teammates with the rat, highlights the epistemic value information seeking component. And then the second half of the chapter kind of like really quickly runs through several related concepts and cites several other papers that deploy discrete models in different cases. Any sort of closing ideas or questions on seven. Yeah, there's a lot of levels of learning and like each of these citations could be investigated and you know modified. >> Yeah, I'd like to echo uh Mark's sentiment. Uh Daniel, you do such a great job of of uh keeping focused on the big picture while explaining the individual parts. I just want to make the comment that I've gone over this and over this so many times and in trying to conceptualize it and it's uh it's it's so helpful to revisit and revisit and revisit. >> Thank you. Yes, it is. It there it is. It is always surprising the the textbook and and other materials like there's just so much to revisit. It's ridiculous. All right. Thank you everyone. See you next time.